INTRODUCTION

The goal of computer vision is to understand the story unfolding in a picture. As humans, this is quite simple. But for computers, the task is extremely difficult.

So why bother learning computer vision?

Well, images are everywhere!

Whether it be personal photo albums on your smartphone, public photos on Facebook, or videos on YouTube, we now have more images than ever – and we need methods to analyze, categorize, and quantify the contents of these images.

For example, have you recently tagged a photo of yourself or a friend on Facebook lately? How does Facebook seem to "know" where the faces are in an image?

Facebook has implemented facial recognition algorithms into their website, meaning that they cannot only *find* faces in an image, they can also *identify* whose face it is as well! Facial recognition is an application of computer vision in the real world.

What other types of useful applications of computer vision are there?

Well, we could build representations of our 3D world using public image repositories like Flickr. We could download thousands and thousands of pictures of Manhattan,taken by citizens with their smartphones and cameras, and then analyze them and organize them to construct a 3D representation of the city. We would then virtually navigate this city through our computers. Sound cool?

Another popular application of computer vision is surveillance.

While surveillance tends to have a negative connotation of sorts, there are many different types. One type of surveillance is related to analyzing security videos, looking for possible suspects after a robbery.

But a different type of surveillance can be seen in the retail world. Department stores can use calibrated cameras to track how you walk through their stores and which kiosks you stop at.

On your last visit to your favorite clothing retailer, did you stop to examine the spring's latest jeans trends? How long did you look at the jeans? What was your facial expression as you looked at the jeans? Did you then pick up a pair and head to the dressing room? These are all types of questions that computer vision surveillance systems can answer.

Computer vision can also be applied to the medical field. A year ago, I consulted with the National Cancer Institute to develop methods to automatically analyze breast histology images for cancer risk factors. Normally, a task like

this would require a trained pathologist with years of experience – and it would be extremely time consuming!

Our research demonstrated that computer vision algorithms could be applied to these images and could automatically analyze and quantify cellular structures – without human intervention! Now, we can analyze breast histology images for cancer risk factors much faster. Of course, computer vision can also be applied to other areas of the medical field. Analyzing X-rays, MRI scans, and cellular structures all can be performed using computer vision algorithms.

Perhaps the biggest success computer vision success story you may have heard of is the X-Box 360 Kinect. The Kinect can use a stereo camera to understand the depth of an image, allowing it to classify and recognize human poses, with the help of some machine learning, of course. The list doesn't stop there.

Computer vision is now prevalent in many areas of your life, whether you realize it or not. We apply computer vision algorithms to analyze movies, football games, hand gesture recognition (for sign language), license plates (just in case you were driving too fast), medicine, surgery, military, and retail. We even use computer visions in space! NASA's Mars Rover includes capabilities to model the terrain of the planet detect obstacles in its path, and stitch together panoramic images.